

Algebra and graphs

1. Solve and explain

1. Read the equation carefully.
Show the same operation on both sides.
Check your answer by substitution.
2. Solve $\log_{10}(4x + 968) = 3$. State the domain restriction.

3. Solve $\cos(x) = -\frac{1}{2}$ for $0 \leq x < 360^\circ$.

Solutions

2. Domain: $4x + 968 > 0$.
Rewrite: $4x + 968 = 10^3 = 1000$.
 $x = 8$. Check: the logarithm argument is $1000 > 0$.
3. $x = 120^\circ, 240^\circ$. Use the unit circle; 360 degrees is excluded.

From formula to graph

2. Quadratic functions

Interpret the graph

$$x^2 - 4$$

4. Identify the roots of $x^2 - 4 = 0$.

Then explain why the minimum occurs at $x = 0$.

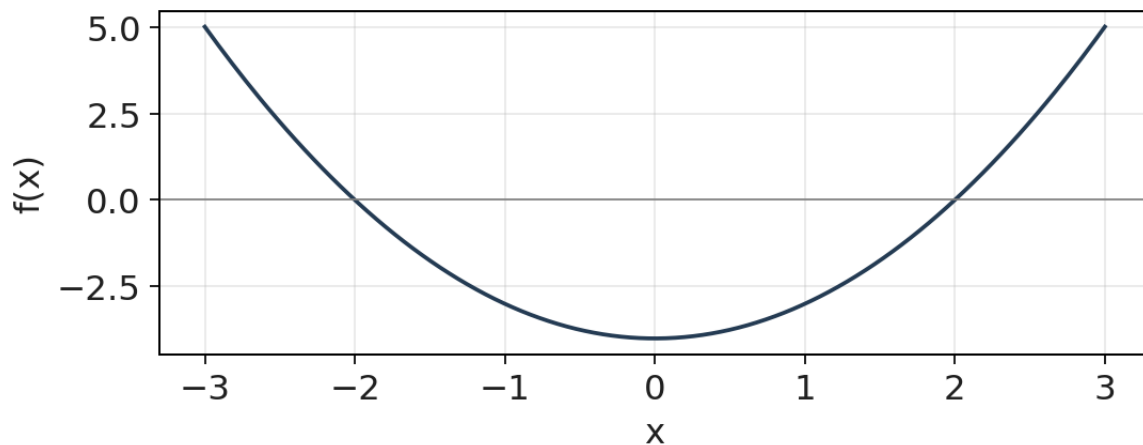


Figure 1. A quadratic with two real roots.

Solutions

4. Roots: $x = -2, 2$.
Minimum: $(0, -4)$.

Notation and line height

Tall formulas remain readable

5. Simplify $\frac{1}{1 + \frac{1}{x}}$.

Explain the domain restriction before cancelling terms.
